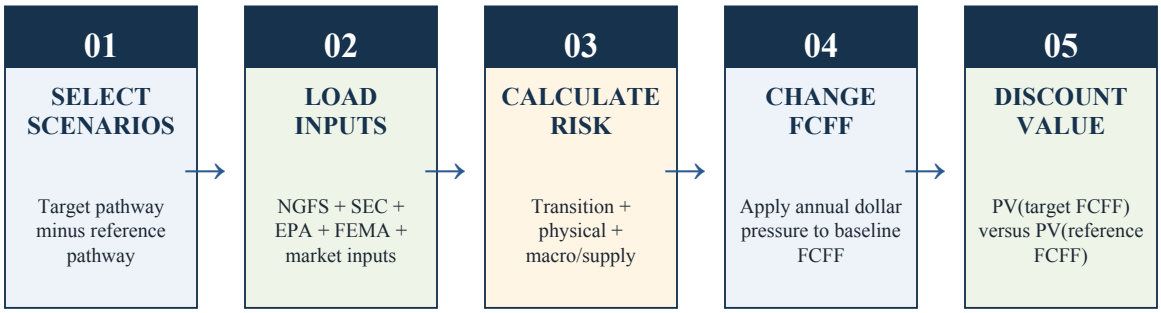
Climate Scenario-to-Value Calculation Workflow

How scenario data becomes transition, physical and macro/supply pressure, then FCFF and discounted valuation



Core output

$$\text{Climate value impact} = [EV(\text{target scenario}) / EV(\text{reference scenario})] - 1 \quad (1)$$

DUK Base example. Current Policies relative to a climate-agnostic baseline produces transition pressure of $30.3m, physical pressure of $79.0m and macro/supply pressure of $49.4m at 2050. After discounting the complete FCFF paths, enterprise value changes from $50.132bn to $47.758bn, or -4.7356%.

Reading rule. The three risk pillars are kept separate until the FCFF step. This prevents a physical loss, carbon cost and macro shock from being hidden inside one unexplained score.

1. Direct transition risk

The model converts policy-sensitive emissions, energy and transition investment into annual FCFF pressure. The target pathway is always compared with the selected reference pathway.

Net transition pressure(t) = net cost(target,t) - net cost(reference,t); pass-through is applied once inside each net cost	(2)
Scope 1 carbon cost(t) = emissions(2023) x emissions pathway(t) x carbon price(t) x incidence x [1 - pass-through]	(3)
Scope 2/3 cost(t) = purchased-emissions base x pathway(t) x carbon price(t) x coverage x incidence x [1 - pass-through]	(4)

DUK worked calculation at 2050

Step	Calculation	DUK result
Starting emissions	EPA 2023 Scope 1 emissions are projected along the scenario-specific emissions path.	72.0 MtCO2e starting point
Carbon price	Use the 2050 difference between Current Policies and the climate-agnostic reference.	$2.1579/tCO2e differential
Gross policy cost	72.0m tCO2e x emissions-path factor x $2.1579/tCO2e.	Carbon-cost base before incidence
Economic incidence	Apply assumed policy incidence and customer pass-through once. Prototype demonstration assumption; not issuer-observed.	Scope 1 net cost = $30.1m
Other channels	Add purchased-energy and value-chain costs using disclosed incidence assumptions. Prototype demonstration assumptions.	Scope 2 = $48k; Scope 3 = $70k
Transition pillar	Sum the target-minus-reference annual costs.	2050 FCFF pressure = $30.3m

Interpretation: $30.3m is an annual cash-flow pressure in the selected scenario comparison. It is not a 30.3% risk score. Its investment meaning appears only after it is incorporated into the multi-year FCFF path and discounted.

2. Physical risk

Physical pressure is calculated in two parts. Acute risk represents event damage to exposed assets. Chronic risk represents gradual productivity or operating pressure on cash flow.

Acute loss(t) = exposed asset value x hazard damage rate(t) x [1 - resilience] x vulnerability	(5)
Chronic loss(t) = reference FCFF(t) x operating exposure x chronic sensitivity x chronic hazard change(t)	(6)
Physical pressure(t) = acute loss(t) + chronic loss(t)	(7)

DUK worked calculation at 2050

Step	Calculation	DUK result
Exposed assets	SEC property, plant and equipment x assumed covered share. The 65% share is a prototype demonstration assumption.	$129.959bn x 65%
Acute hazard	Facility/county FEMA context is converted using an assumed scenario factor. The financial damage conversion is not observed.	0.17566% modeled damage-rate change
Protection	Reduce gross damage for assumed resilience and vulnerability. Prototype demonstration assumptions.	(1 - 50%) x 1.00
Acute loss	$129.959bn x 65% x 0.17566% x (1 - 50%) x 1.00.	$74.2m
Chronic loss	$2.513bn reference terminal FCFF x 10% exposure x 25% conversion x 7.57987% hazard change. The 10% and 25% are demonstration assumptions.	$4.8m
Physical pillar	$74.2m acute + $4.8m chronic.	2050 FCFF pressure = $79.0m

Interpretation: the physical pillar is a dollar loss to operations and assets. It enters valuation by lowering scenario FCFF, rather than by adding an arbitrary percentage directly to enterprise value.

3. Macro and supply-chain risk

The model transmits scenario GDP and supply-chain changes into company cash flow. This captures the indirect effect of climate pathways on demand, input availability and operating costs.

Macro/supply rate(t) = [GDP shock(t) + supply shock(t)] x sector elasticity x issuer scale	(8)
Issuer scale = clip[0.50 + CapEx / operating cash flow, 0.50, 2.00]	(9)
Macro/supply pressure(t) = reference FCFF(t) x macro/supply rate(t)	(10)

DUK worked calculation at 2050

Step	Calculation	DUK result
Scenario shock	Read Current Policies GDP and supply-chain difference relative to the reference.	Combined terminal shock = 2.00%
Sector transmission	Apply assumed Utilities sensitivity to a common macro/supply shock. Prototype demonstration assumption.	Sector elasticity = 0.60
Issuer scale	0.50 + CapEx / operating cash flow, bounded to [0.50, 2.00]. The transformation rule is a demonstration assumption.	DUK scale = 1.6374
FCFF rate	2.00% x 0.60 x 1.6374.	Terminal pressure rate = 1.9649%
Macro/supply pillar	$2.513bn reference terminal FCFF x 1.9649%.	2050 FCFF pressure = $49.4m

Interpretation: this pillar is not direct carbon cost or direct storm damage. It represents the indirect change in company cash flow transmitted through the economy, sector demand and supply chain.

4. From the three risk pillars to FCFF

Each scenario has a complete annual FCFF path. The model starts from normalized company FCFF, applies baseline growth, and subtracts the target-minus-reference pressure from each pillar.

Normalized unlevered-FCFF approximation = operating cash flow - capital expenditure + after-tax interest	(11)
Scenario FCFF(t) = reference FCFF(t) - transition pressure(t) - physical pressure(t) - macro/supply pressure(t)	(12)
WACC = E/(D+E) x [risk-free rate + beta x equity risk premium] + D/(D+E) x debt cost x (1-tax rate)	(13)
EV(s) = sum from t=1 to T of FCFF_s(t)/(1+WACC)^t + [FCFF_s(T+1)/(WACC-g)]/(1+WACC)^T	(14)

DUK valuation bridge

Step	Calculation	DUK result
2025 baseline	SEC operating cash flow - capital expenditure + after-tax interest. This is an internally calculated unlevered-FCFF approximation.	Normalized FCFF = $1.532bn
Reference path	Grow normalized FCFF using the disclosed annual growth assumption. This is a prototype demonstration path, not an issuer forecast.	2050 reference FCFF = $2.513bn
Subtract risks	$2.513bn - $30.3m - $79.0m - $49.4m.	2050 target FCFF = $2.354bn
Discount rate	Market/SEC inputs plus assumed equity premium and terminal growth. ERP and terminal growth are demonstration assumptions.	WACC = 5.116%; terminal growth = 2.00%
Reference value	Discount reference FCFF path and terminal value.	Reference EV = $50.132bn
Target value	Discount target FCFF path and terminal value.	Target EV = $47.758bn
Value impact	$47.758bn / $50.132bn - 1.	-4.7356%

Investment output: under Current Policies relative to the climate-agnostic baseline, the prototype produces a 4.74% lower enterprise-value sensitivity for DUK. It is not an equity value, share-price forecast or audited valuation; the non-observed conversions above are demonstration assumptions.

5. DUK end-to-end calculation summary

This page is the complete calculation chain in one view.

Stage	Calculation	Output
A. Scenario comparison	Current Policies - climate-agnostic baseline	Defines every target-minus-reference shock
B. Transition	EPA emissions + scenario carbon price + energy/CapEx/pass-through	$30.3m annual FCFF pressure at 2050
C. Physical	SEC exposed assets + FEMA hazard damage + resilience; chronic operating exposure	$79.0m annual FCFF pressure at 2050
D. Macro / supply	NGFS GDP/supply shock x Utilities elasticity x DUK scale	$49.4m annual FCFF pressure at 2050
E. Scenario FCFF	$2.513bn - $30.3m - $79.0m - $49.4m	$2.354bn target terminal FCFF
F. Discounted values	Discount both annual FCFF paths at 5.116%, with 2.00% terminal growth	$50.132bn reference EV; $47.758bn target EV
G. Climate value impact	$47.758bn / $50.132bn - 1	-4.7356%

Final risk-to-value attribution: transition -0.8954% + physical -2.4384% + macro/supply -1.4018% = total -4.7356% (15)

What the investment team should use: compare only RANKABLE issuers within the same scenario and assumption case, then inspect the three pillar contributions. Proxy, floor or denominator-sensitive results are excluded from ordinary ranking.

Primary inputs used in this calculation: SEC company financial statements for FCFF and asset values; EPA emissions records; FEMA facility/county hazard context; NGFS scenario pathways for macroeconomic, transition and physical variables; and market inputs for WACC.